

The Haldane Model with a Single $+\pi/3$ Disclination: the explicit relational data, step by step

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Abstract

This note instantiates the relational substrate of [*A Relational Observable Substrate and its NCG Realization*] on one system: the Haldane model carrying a single $+\pi/3$ disclination. Every object of the pipeline — contexts, the category of comparisons and its groupoid core, the computed nerve, the universal nest algebra and its representations, Fermi projections, the universal calculus, the cone vertex and its extension problem, transport determinants — is written explicitly for this lattice. Steps whose general construction is not yet established are marked **Open**, each with the intended method when one exists.

1 The physical system

Definition 1.1 (Defect-free data). *Let Λ_0 be the honeycomb lattice with lattice constant a , sublattices A, B , nearest-neighbour vectors $\delta_1, \delta_2, \delta_3$, and $\mathcal{H}_0 = \ell^2(\Lambda_0)$. The Haldane Hamiltonian is*

$$H = t_1 \sum_{\langle x, y \rangle} |x\rangle\langle y| + t_2 \sum_{\langle\langle x, y \rangle\rangle} e^{i\varphi \nu_{xy}} |x\rangle\langle y| + M \sum_x \epsilon_x |x\rangle\langle x|, \quad (1)$$

with $\epsilon_x = +1$ on A , -1 on B , and $\nu_{xy} = \text{sgn}(\mathbf{d}_1 \times \mathbf{d}_2)_z = \pm 1$ the standard chirality of the next-nearest-neighbour hop $x \rightarrow y$ through the intermediate site. Parameters are fixed in the topological regime

$$|M| < 3\sqrt{3} |t_2 \sin \varphi|, \quad C = \text{sgn}(\varphi) = \pm 1. \quad (2)$$

Definition 1.2 (Disclinated lattice). *Fix a plaquette centre d_0 of Λ_0 and the six closed 60° sectors $W_1, \dots, W_6$ at d_0 bounded by consecutive rays through the plaquette's vertices. The disclinated lattice Λ is the graph obtained from the five sectors $W_1, \dots, W_5$ by identifying, for each $k \in \mathbb{Z}/5$, the right boundary ray of W_k with the left boundary ray of W_{k+1} site-by-site (indices mod 5). The core plaquette of Λ is a pentagon; every other plaquette is a hexagon; the Frank angle is $\Omega = +\pi/3$. The one-particle space is $\mathcal{H}_\Lambda = \ell^2(\Lambda)$; the sublattice labels A, B and the couplings (t_1, t_2, φ, M) of (1) carry over edge-by-edge. We write $o \in \Lambda$ -plane for the core point (the pentagon's centre); o is a point of the lattice plane, not a datum of the formalism.*

Remark 1.3. *Away from the core the chirality ν_{xy} is determined by (1) in any local chart. On the five next-nearest-neighbour bonds surrounding the pentagon the six-fold rule leaves a finite ambiguity; ν is fixed there by the chart of the context containing the bond (Definition 2.2), a gauge choice affecting no statement below.*

2 Contexts, the comparison category, and the computed nerve

Definition 2.1 (Contexts). *A context is a finite region $U \subset \Lambda$ admitting a chart: an injective map $\kappa : U \rightarrow \Lambda_0$ preserving edges, sublattice labels, and next-nearest-neighbour chiralities. A region containing the pentagon admits no chart; the defect is an absence of context, not a*

distinguished context. For the star cover, fix a collar width $w \in \mathbb{N}$ and an inner radius r_0 large enough that only angularly adjacent sectors meet at distance $\geq r_0$ from o , and set

$$U_i = \{x \in \Lambda : (x \in W_i \text{ or } \text{dist}(x, \partial W_i) \leq w) \text{ and } \text{dist}(x, o) \geq r_0\}, \quad i \in \mathbb{Z}/5,$$

the annular thickened sectors, with overlaps $U_{i,i+1} = U_i \cap U_{i+1}$ (radial strips) and $U_i \cap U_j = \emptyset$ for $j \neq i \pm 1$.

Definition 2.2 (Patch charts). For each i fix a chart $\kappa_i : U_i \rightarrow \Lambda_0$; it exists because each annular sector is disclination-free, and is unique up to a symmetry of Λ_0 . On an overlap, two charts differ by a rigid motion of the honeycomb:

$$\kappa_{i+1} \circ \kappa_i^{-1} \Big|_{\kappa_i(U_{i,i+1})} = m_{i,i+1} \in \text{Sym}(\Lambda_0), \quad m_{5,1} m_{4,5} m_{3,4} m_{2,3} m_{1,2} = R(\pi/3), \quad (3)$$

the composition around the core being the Frank-wedge rotation — this is Definition 1.2 read through the charts.

Definition 2.3 (The comparison category and its core). $\mathcal{C}_Q$ is the category whose objects are the contexts and overlaps present in the data, and whose arrows are generated by the restrictions $\rho_{i,i+1}^i : \mathcal{A}_i \rightarrow \mathcal{A}_{i,i+1}$ (Definition 3.2, not invertible) and the comparisons $\alpha_{i,i+1}$ (Definition 3.3, invertible), with the relations they satisfy in the data and no others. The relational groupoid is its invertible part,

$$R_Q = \text{Core}(\mathcal{C}_Q),$$

generated here by the arrows $\gamma_{i,i+1} : i \rightarrow i+1$ underlying the $\alpha_{i,i+1}$. No relation is imposed on the composites of the $\gamma_{i,i+1}$: in particular the pentagon loop $\gamma_{5,1}\gamma_{4,5}\gamma_{3,4}\gamma_{2,3}\gamma_{1,2}$ is not declared trivial, nor declared nontrivial; whatever it is, the calculus below must detect it.

Definition 2.4 (Nerve and incidence rule). The nerve $K = N(R_Q)$ is the simplicial set with p -simplexes the composable p -tuples of arrows of R_Q ; equivalently, on vertex sets,

$$[i_0, \dots, i_p] \in K \iff \text{a composable chain of comparisons between these contexts is present in the data.}$$

The simplexes of K are computed from the data by this rule; only the graphical placement of vertices is free.

Observation 2.5 (The computed nerve of the star). For the cover of Definition 2.1 the comparisons present in the data are exactly the five $\alpha_{i,i+1}$ on adjacent annular overlaps; non-adjacent contexts share no overlap and no comparison. Hence

$$K = C_5 \quad (\text{five vertices, five edges } \{i, i+1\}, \text{ no 2-simplexes}).$$

The same rule applied to six annular sectors around a bulk plaquette of Λ_0 yields the reference nerve $K^{\text{ref}} = C_6$. The two crystals produce different sites; neither is postulated.

Open — dependence on the incidence rule. Candidate rules ($U_{i_0 \dots i_p} \neq \emptyset$ set-theoretically; a common algebra $\mathcal{A}_{i_0 \dots i_p} \neq 0$; a composable chain of comparisons) coincide for this cover but not in general. Whether the classes of §7–9 are independent of the rule is not established; no candidate proof yet.

3 Local algebras, restrictions, comparisons

Definition 3.1 (Local observable algebras). *For each context and overlap,*

$$\mathcal{H}_i = \ell^2(U_i) \cong \mathbb{C}^{N_i}, \quad \mathcal{A}_i = B(\mathcal{H}_i) = M_{N_i}(\mathbb{C}), \quad \mathcal{A}_{i,i+1} = B(\ell^2(U_{i,i+1})),$$

with $N_i = \#U_i$; the matrix units $e_{xy} = |x\rangle\langle y|$, $x, y \in U_i$, generate $\mathcal{A}_i$.

Definition 3.2 (Restriction maps). *With $\chi_{i,i+1} \in \mathcal{A}_i$ the orthogonal projection onto $\ell^2(U_{i,i+1})$,*

$$\rho_{i,i+1}^i(a) = \chi_{i,i+1} a \chi_{i,i+1} \in \mathcal{A}_{i,i+1},$$

unital completely positive, and multiplicative on the subalgebra of operators supported in $U_{i,i+1}$; all descent conditions below involve only such elements.

Definition 3.3 (Comparison isomorphisms). *Let $S_{i,i+1} : \ell^2(U_{i,i+1}) \rightarrow \ell^2(U_{i+1})$ be the unitary induced by the change of chart: it acts as the identity on the physical sites and transports the frame data (chart-dependent labels, chirality conventions) by $m_{i,i+1}$ of (3). The comparison is*

$$\alpha_{i,i+1} = \text{Ad}(S_{i,i+1}) : \mathcal{A}_i|_{i,i+1} \xrightarrow{\sim} \mathcal{A}_{i+1}|_{i,i+1}.$$

On any finer cover of the bulk of Λ whose computed nerve contains 2-simplexes, the induced comparisons compose strictly there, $\alpha_{ij}\alpha_{jk} = \alpha_{ik}$; on the star cover no 2-simplex exists (Observation 2.5) and no composition relation is imposed.

4 The universal nest algebra and realization

Definition 4.1 (Universal nest algebra). *The universal algebra of the nest is the quotient of a free product by the nest's own relations,*

$$\mathcal{A}_u = \left(\bigast_{i \in \mathbb{Z}/5} \mathcal{A}_i \right) * \mathbb{C} \langle v_{i,i+1}^{\pm 1} : i \in \mathbb{Z}/5 \rangle \Big/ \left\langle v_{i,i+1} \rho_{i,i+1}^i(a) v_{i,i+1}^{-1} = \rho_{i,i+1}^{i+1}(\alpha_{i,i+1}(a)), v_{jk} v_{ij} = v_{ik} \text{ on 2-simplexes} \right\rangle$$

Every $\mathcal{A}_i$ embeds; every comparison becomes inner. Since $K = C_5$ has no 2-simplexes, no composition relation constrains the $v_{i,i+1}$ here: the pentagon word

$$w_d = v_{5,1} v_{4,5} v_{3,4} v_{2,3} v_{1,2} \in \mathcal{A}_u$$

is unconstrained by the defining relations. It is the algebraic carrier of whatever holonomy the crystal has; its value is fixed only by a representation.

Definition 4.2 (Realization as representation). *By the universal property of $\mathcal{A}_u$,*

$$\text{Rep}(\mathcal{A}_u, \mathcal{H}) \cong \left\{ (\pi_i, V_{i,i+1}) : V_{i,i+1} \pi_i(a) V_{i,i+1}^* = \pi_{i+1}(\alpha_{i,i+1}(a)) \text{ on overlaps} \right\},$$

so a compatible family of local representations and implementers is a representation of $\mathcal{A}_u$. The concrete representation used below is

$$\pi_i = \text{id} : \mathcal{A}_i \rightarrow B(\mathcal{H}_i), \quad V_{i,i+1} = S_{i,i+1} \oplus \mathbf{1} \quad \text{on} \quad \ell^2(U_{i,i+1}) \oplus \ell^2(U_{i,i+1})^\perp,$$

each $V_{i,i+1}$ determined by $\alpha_{i,i+1}$ up to a unitary of the commutant; the determinant phases of §9 are insensitive to this ambiguity after the reference ratio.

Definition 4.3 (Local traces). *On each $\mathcal{A}_i = M_{N_i}(\mathbb{C})$ take the normalized trace $\text{tr}_i = \frac{1}{N_i} \text{Tr}$.*

Open — completion and faithfulness of $\mathcal{A}_u$. $\mathcal{A}_u$ is a bare $*$ -algebra; whether a universal C^* -completion is non-degenerate, whether it carries faithful representations for a general nest, and whether $K_0(\mathcal{A}_u)$ is the home of the projections of §5, is not established. Intended method: sup over the representations of Definition 4.2.

Open — the global trace. A faithful trace $\mathcal{T}$ on the realized algebra must restrict compatibly to the tr_i across overlaps. Intended method: define $\mathcal{T}$ simplex-wise on K with inclusion-exclusion weights, and verify compatibility as the degree-0 part of the graph-generated cochains of §8. Not yet established.

5 Sector selection

Definition 5.1 (Local Hamiltonians and Fermi projections). $H_\Lambda \in B(\mathcal{H}_\Lambda)$ is (1) summed over the edges of Λ (chiralities fixed by Remark 1.3). For each context,

$$H_i = \chi_{U_i} H_\Lambda \chi_{U_i} \in \mathcal{A}_i, \quad P_i = \chi_{(-\infty, E_F)}(H_i), \quad E_i = \text{Ran } P_i,$$

with E_F in the bulk spectral gap of (2).

Open — boundary spectrum in the gap. The truncation H_i carries edge states of ∂U_i inside the bulk gap, so P_i depends on where E_F sits among them. Intended method: define P_i instead as the compression $\chi_{U_i} P_\Lambda \chi_{U_i}$ of the full Fermi projection, followed by spectral flattening ($P_i = \chi_{(1/2, 1]}$ of the compression); which choice makes the identities of §6–§9 exact is not settled.

6 The universal calculus, connection, curvature

Definition 6.1 (Universal differential calculus). For a unital algebra $\mathcal{A}$, $\Omega_u^\bullet(\mathcal{A})$ is the differential graded algebra generated by $\mathcal{A}$ in degree 0 and symbols da ($a \in \mathcal{A}$) in degree 1, subject only to linearity of d , $d1 = 0$, and the Leibniz rule $d(ab) = da \cdot b + a \cdot db$; thus $\Omega_u^n(\mathcal{A})$ is spanned by $a_0 da_1 \cdots da_n$, and $da = 1 \otimes a - a \otimes 1$ under the identification $\Omega_u^n(\mathcal{A}) \cong \mathcal{A} \otimes \bar{\mathcal{A}}^{\otimes n}$. It is purely algebraic and functorial: every comparison $\alpha_{i,i+1}$ lifts canonically to $\Omega_u(\alpha_{i,i+1})$ intertwining d , so the calculus is carried by the nest with no metric injected.

Definition 6.2 (Connection and curvature). On the realized algebra,

$$a_{i,i+1} = dV_{i,i+1}; \quad F_{ijk}^{\text{rel}} = P_i dV_{ij} dV_{jk} \quad \text{one element per 2-simplex } \{i, j, k\} \text{ of the computed nerve.}$$

On the star cover, $K = C_5$ has no 2-simplexes, so the curvature 2-cochain on K is void: the curvature content of the defect migrates to the cone of §7. On finer bulk covers with genuine 2-simplexes, F^{rel} is the 2-cochain paired by the characteristic map of §8.

Observation 6.3 (Closure). From $P_i^2 = P_i$: $P_i(dP_i)P_i = 0$, hence $\nabla F = 0$ for $\nabla = P d$. On bulk 2-simplexes, strict composition and Leibniz give $dV_{ik} = dV_{ij} V_{jk} + V_{ij} dV_{jk}$, hence $\delta F^{\text{rel}} = 0$ there. Both are one-line calculus identities.

7 The cone vertex and the extension problem

Definition 7.1 (The cone vertex). Adjoin to $\mathcal{C}_Q$ a freely added terminal object d : no algebra $\mathcal{A}_d$, no context U_d , no operator data are attached to d . On nerves this is the simplicial cone

$$C_d K = d * K, \quad \sigma = \{i_0, \dots, i_p\} \in K \mapsto d * \sigma = \{d, i_0, \dots, i_p\} \in C_d K.$$

d is central by construction — it is adjacent to every context in the referential sense, not the metric one — and it is the same formal vertex for every crystal; what depends on the crystal is the extension problem below.

Definition 7.2 (The extension problem). Given a cochain $c \in C^p(K; \mathcal{V})$ built from the physical data, the question attached to d is whether c extends compatibly over the cone:

$$\exists ? \tilde{c} \in C^p(C_d K; \mathcal{V}), \quad \tilde{c}|_K = c, \quad D\tilde{c} = 0,$$

D the total (Čech–NCG) differential. The vertex adds a constraint, not a facilitation: it asks whether the pairwise comparisons of K participate in a single global structure. Since $C_d K$ is contractible, the content of the obstruction is relative:

$$\check{H}^{p+1}(C_d K, K) \cong \check{H}^p(K), \quad \text{here} \quad \check{H}^2(C_d K, K; U(1)) \cong \check{H}^1(C_5; U(1)). \quad (4)$$

The 2-simplexes $\{d, i, i+1\}$ of the cone carry extension data, not physical data.

Open — the coefficient complex. In (4) the coefficients are taken in $U(1)$ (transport determinants, §9). The full receptacle should be the total cohomology of a Čech–NCG bicomplex $\check{C}^p(K; CC_q)$ with cyclic-chain coefficients, in which the extension problem of Definition 7.2 is posed for the total differential and a total connection whose $(1, 1)$ -component measures $P_{i+1} - V_{i,i+1} P_i V_{i,i+1}^*$. No part of this bicomplex has been constructed yet; no candidate method is fixed.

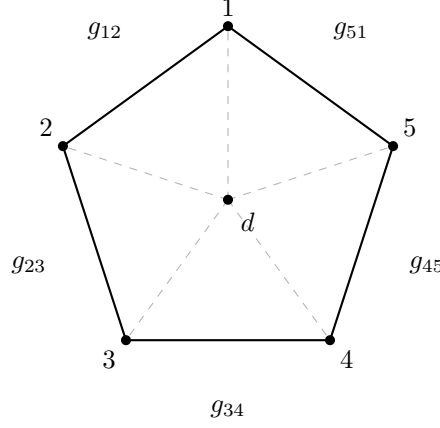


Figure 1. The cone $C_d K = d * K$. Solid outer cycle: the computed nerve $K = C_5$ (Observation 2.5), carrying the physical data $g_{i,i+1}$. Dashed: the cone edges $\{d, i\}$ and triangles $\{d, i, i+1\}$, carrying extension data (Definition 7.2). The pentagon is not a piece of the nerve; it is the augmentation testing whether the nerve’s data cohere globally.

8 The characteristic map

Definition 8.1 (Area cocycle, given the derivations). For derivations ∂_1, ∂_2 on the realized algebra and the trace $\mathcal{T}$, the Connes–Chern (area) cocycle is

$$\tau(a_0, a_1, a_2) = \frac{1}{2\pi i} \mathcal{T} \left(a_0 (\partial_1 a_1 \partial_2 a_2 - \partial_2 a_1 \partial_1 a_2) \right), \quad c_{ijk} = \tau(F_{ijk}^{\text{rel}}),$$

defined on the 2-simplexes of bulk covers; the bulk class is $c_1^{\text{rel}} = [c] \in \check{H}^2$, integralized by the index pairing $\langle \tau, [P] \rangle$.

Open — the position cocycle. On Λ there is no global position: the derivations must come from a groupoid 1-cocycle $X : R_Q \rightarrow \mathbb{R}^2$, $X(\gamma\gamma') = X(\gamma) + \gamma \cdot X(\gamma')$, with $\partial_j = [X_j, \cdot]$. Around the pentagon the closure constraint involves the Frank rotation. Intended method: define X on the generators $\gamma_{i,i+1}$ from the patch charts κ_i and absorb $R(\pi/3)$ into a $\mathbb{Z}_6$ -action, i.e. a cocycle valued in $\mathbb{R}^2 \rtimes \mathbb{Z}_6$. Construction attempted, not verified.

Open — the cochains via graph cohomology. The cocycle τ (and its higher companions) should be generated, not chosen: a chain map $\Gamma \mapsto \tau_\Gamma$ from a graph complex to cyclic cochains, with purely combinatorial weights, following the LRRH pattern. Existence of this map at the C^* -level is not established; it is the intended source of the $\mathcal{T}$ -compatible cochains of §4.

9 Defect readout

Definition 9.1 (Transport cochain). For each edge of K ,

$$g_{i,i+1} = \frac{\det(V_{i,i+1}|_{E_i})}{|\det(V_{i,i+1}|_{E_i})|} \in U(1),$$

where $V_{i,i+1}|_{E_i}$ is the compression $P_{i+1}V_{i,i+1}P_i$ regarded as a map $E_i \rightarrow E_{i+1}$ (frames fixed once per context). Since $K = C_5$ has no 2-simplexes, $g \in C^1(K; U(1))$ is automatically a cocycle.

Definition 9.2 (Obstruction cochain on the cone). Choose any extension of g to the cone: phases $h_i \in U(1)$ on the edges $\{d, i\}$. The failure of D -closure on the cone triangles is the relative 2-cochain

$$\omega_{d,i,i+1} = h_{i+1}^{-1} g_{i,i+1} h_i, \quad \prod_{i \in \mathbb{Z}/5} \omega_{d,i,i+1} = \prod_{i \in \mathbb{Z}/5} g_{i,i+1} =: \Phi_d,$$

independently of the h_i . An extension with $\omega \equiv 1$ exists iff $\Phi_d = 1$; the class $[\omega] \in \check{H}^2(C_d K, K; U(1)) \cong U(1)$ has value Φ_d . The vertex d converts the global-coherence question on K into a relative obstruction on the cone.

Definition 9.3 (Reference and decision). The reference system is a bulk plaquette of Λ_0 with six annular contexts, same collar and inner radius, same Definitions 2.1–9.2 with $K^{\text{ref}} = C_6$; this yields Φ_d^{ref} . The decision is

$$\Delta\Phi = \arg \frac{\Phi_d}{\Phi_d^{\text{ref}}} \in \mathbb{R}/2\pi\mathbb{Z}, \quad c_d^{\text{rel}} \neq 0 \iff \Delta\Phi \notin 2\pi\mathbb{Z},$$

with the integer lift obtained by branch-consistent logarithms of the $g_{i,i+1}$ relative to the reference. $\Delta\Phi$ is built from the transport determinants alone: it requires the implementers and the Fermi projections, not the derivations of §8.

Open — the value. The expected form is $\Delta\Phi = \frac{\Omega}{2\pi} \cdot \Theta_{\text{band}} \pmod{2\pi}$ with $\Omega = \pi/3$ the Frank angle and Θ_{band} a bulk band datum of (1) (the comparison point is the Teo–Kane defect classification). Neither the formula nor the constant is derived; the intended check is numerical evaluation of Definitions 9.1–9.3 on Λ at fixed (t_1, t_2, φ, M) in the regime (2).

10 Status of each step

Object	Status	Where
Lattice Λ , contexts U_i , charts κ_i	explicit	§1–2
Category $\mathcal{C}_Q$, groupoid $R_Q = \text{Core}(\mathcal{C}_Q)$, incidence rule	explicit	§2
Computed nerves $K = C_5$, $K^{\text{ref}} = C_6$	explicit (computed, not posed)	§2
Dependence on the incidence rule	open (no method yet)	§2
$\mathcal{A}_i$, ρ , $\alpha_{i,i+1}$, bulk strictness	explicit	§3
Universal nest algebra $\mathcal{A}_u$, pentagon word w_d	explicit as $*$ -algebra	§4
C^* -completion of $\mathcal{A}_u$	open (method sketched)	§4
Realization $(\pi_i, V_{i,i+1})$, local traces	explicit	§4
Global trace $\mathcal{T}$	open (method sketched)	§4
H_i , P_i , E_i	explicit up to boundary choice	§5
Universal calculus Ω_u , $a_{i,i+1}$, F^{rel} , closure	explicit	§6
Cone vertex d , extension problem, relative iso	explicit	§7
Čech–NCG bicomplex, $(1, 1)$ -component	open (no method yet)	§7
Position cocycle X , derivations ∂_j	open (candidate method)	§8
Graph-generated cochains	open (intended route)	§8
$g_{i,i+1}$, ω , Φ_d , $\Delta\Phi$, decision	explicit	§9
Value of $\Delta\Phi$	open (numerical check intended)	§9